

Employment barriers experienced at different job acquisition stages by people with and without disabilities

Erica S. Jablonski*, Kimberly G. Phillips and Megan Henly
Institute on Disability, University of New Hampshire, Durham, NH, USA

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Abstract.

BACKGROUND: Employment barriers experienced by people with disabilities are well-documented by researchers. A closer look at the nature of these barriers at different stages of the job acquisition process may help to understand and address current challenges.

OBJECTIVE: To understand how employment barriers differ for job seekers with and without disabilities; the impact of disability type; and are job acquisition barriers most salient, when looking for, applying to, or accepting a position.

METHODS: 3,021 working-age adults (including 1,491 adults with disabilities) from a Qualtrics opt-in panel completed an online survey about employment status, job search barriers, and demographic factors (including disability). Logistic regression predicted odds of experiencing a job search barrier based on disability and other sociodemographic factors.

RESULTS: Most job seekers reported barriers to employment, with disability being the strongest predictor. While the types of barriers experienced at all stages of the job search were similar for people with and without disabilities, people with disabilities experienced barriers significantly more often.

CONCLUSION: Employment barriers among people with and without disabilities are similar in nature but vary by degree. Improvements in transportation and better alignment between education, training, and the job market may benefit all job seekers.

Keywords: Employment, people with disabilities, job application, sociodemographic factors

1. Introduction

The percentage of people aged 16 to 64 with a disability who are working is substantially lower (37.7%) than those with no disability (75.5%)

(Kessler Foundation & the University of New Hampshire, 2023b). Several U.S. federal policies have attempted to address access to employment for people with disabilities. This includes the 1973 Rehabilitation Act which provided funding to states for vocational rehabilitation programs and also prohibited federal employers from not hiring people due to their disability. The Americans with Disabilities Act of 1990 expanded these disability discrimination protections to non-federal employers and

*Address for correspondence: Erica S. Jablonski, College of Professional Studies, University of New Hampshire, 88 Commercial Street, Manchester, NH 03101, USA. E-mail: e.jablonski@unh.edu.

also required eligible employers to offer reasonable accommodations to workers as needed. The Workforce Investment Act in 1998 expanded vocational rehabilitation (VR) training and programs at the state level, providing one-stop access for services and identifying opportunities for job search and job placement for job seekers with disabilities. Despite these federal interventions, and although the US has seen recent gains in the employment rate among people with disabilities, a significant employment gap persists between people with and without disabilities (Kessler Foundation & the University of New Hampshire, 2023b). For example, in July of 2023, the labor participation rate differential was similar to the employment rate pattern, with 40.4% of people with disabilities aged 16–64 participating in the workforce, as compared to 78.4% of their contemporaries without disabilities. Historically, lower employment rates among people with disabilities were common, and times of economic upset and recession affected this part of the population longer and more severely than people without disabilities (Kessler Foundation & the University of New Hampshire, 2023a; Livermore & Honeycutt, 2015; McAlpine & Alang, 2021). Prior research has shown that these effects do not reflect a lack of desire or effort to be employed on the part of people with disabilities. In fact, the majority (68.4%) have demonstrated that they are striving to work, meaning that they are either currently employed, held jobs in the past, or had been actively seeking a position (Sundar et al., 2018). Despite the desire and efforts to obtain and maintain work, people with disabilities experience myriad barriers to employment (Khayat-zadeh-Mahani et al., 2020; McKinney & Swartz, 2021; Schur et al., 2017).

Employment barriers faced by people with disabilities was a special topic of the first disability supplement to the monthly Current Population Survey ([CPS], Bureau of the Census, 2012). Findings from the supplemental data collection showed that, among people with disabilities who were unemployed or not in the labor force, about 50% reported at least one barrier to employment (Bureau of Labor Statistics, 2013). The most commonly faced challenge was their own disability, followed by lack of education or training, lack of transportation, and the need for special features at the job (Bureau of Labor Statistics, 2013). Informed largely by these findings, the purpose of this article is to update and expand the current understanding of employment obstacles faced by people with disabilities during the job acquisition process.

We replicated the CPS disability supplement questions about barriers that job seekers commonly face. We also compared the nature and frequency of barriers experienced by people with and without disabilities, and we dove deeper to inquire about the different key stages of job acquisition. Based on prior research and the fact that different subpopulations have varying success with distinct aspects of obtaining employment (Barbulescu, 2015; Dannals et al., 2021; Hernandez et al., 2019), we conceptualized three stages of job acquisition: 1) looking for a job means identifying desirable and potentially-attainable positions; 2) applying for a job includes the cover letter, application, and interview process; and 3) accepting a job includes negotiating pay and work schedule, and requesting any needed accommodations. Specifically, this study sought to answer the following research questions (RQ):

RQ1: What types of barriers are job seekers with disabilities most likely to face, compared to job seekers without disabilities?

RQ2: How do job search barriers differ by type of disability?

RQ3: To what extent are job search barriers more salient when looking for, applying to, or accepting a position?

Barriers to employment can be categorized at the levels of the individual employee, the employing organization, and the larger societal level, as reflected in Table 1 (Sundar et al., 2018).

1.1. Individual-level barriers

Individual-level factors include prospective employees' characteristics (e.g., age, race, gender, education, disability) that may be self-identified as contributing to employment difficulty, such as in a study by Fyffe et al. (2022) in which personal characteristics (e.g., medical conditions, functional limitations) were mentioned more frequently than employer or societal-level barriers to employment among working-age adults with disabilities (Fyffe et al., 2022). Individual-level factors can also be cited by others (e.g., managers, employers) as causes of work limitation. For example, disability type can become significant, when managers perceive more barriers among people with intellectual or mental disabilities compared to people with physical disabilities (Kocman, Fisher, & Weber, 2018).

One other individual-level barrier is navigating a need for accommodation, potentially creating barriers

Table 1
Examples of employment barriers by level

Individual-level barriers
Disability/health condition
Lack of education/training
Need for accommodations (self-identified as a barrier)
Other (e.g., age)
Organizational-level barriers
Employer or coworker attitudes (may include misconceptions, e.g., extra costs for accommodations)
Other (e.g., background check)
Societal-level barriers
Lack of transportation
Loss of benefits or government assistance (e.g., Social Security)
Other (e.g., daycare costs too expensive)

ers at the time of disclosing the need, requesting an accommodation, or receiving the needed accommodation. This problem has emerged in prior qualitative research regarding disability disclosure issues (Inge et al., 2018). In a more recent Canadian study surveying young adults transitioning to employment, disclosure was the most commonly indicated job accommodation barrier (77%), followed by perceived cost of accommodation (74%), and potential inability to accommodate job duties (68%) (Jetha et al., 2019). Individual-level barriers also include a lack of training or education that would facilitate employment. Among recent college graduates in the US, it was not disability, but lack of training that was identified as the most prevalent barrier to employment for job seekers with and without disabilities (Phillips et al., 2020). In similar research findings from Australia, people with intellectual disabilities described not receiving the skills and knowledge that could help them secure a position of interest from employment services (Meltzer, Robinson, & Fisher, 2020).

1.2. Organizational-level barriers

Organizational-level factors that serve as barriers to employment for people with disabilities include employer perceptions and misconceptions. For example, in an environmental scan of publications on employer policies and practices conducted by Baker and his colleagues (2018) several studies identified instances where employers expressed concerns about the sufficiency of educational and training credentials or qualifications of applicants with disabilities. People with disabilities also report employer beliefs that they were not qualified (Houtenville et al., 2015; Sundar et al., 2018). While this was found to be the case among employing organizations as well, employer reports did not directly speak to non-inclusive work-

place cultures that people with disabilities reported as a barrier to employment (Chen, 2015; Meltzer, Robinson, & Fisher, 2020). Employers did acknowledge having greater hiring concerns for people with intellectual disabilities or mental disorders than for those with physical disabilities (Kocman, Fischer, & Weber, 2018), consistent with the experiences of stigma and discrimination described by employees with intellectual disabilities in competitive workplaces (Meltzer, Robinson, & Fisher, 2020), a group that has reported discrimination as their greatest barrier to employment (Jacob et al., 2022).

Doubt or questions about applicant capacity also emerged as a barrier in a qualitative study of South Africans engaged in the job search process (McKinney & Swartz, 2021), even when those capabilities were not clearly associated with the responsibilities of a position. In Austria, 30 hiring managers emphasized a lack of appropriate jobs, frequently relating this to high organizational job requirements (Kocman, Fischer, & Weber, 2018). The purely structural argument in that study seemed undermined however by the association found between perceived employment barriers and manager attitudes. While perceptions about people with intellectual disabilities were not said to influence beliefs about their employability, managers who saw them as more different from people without disabilities, or who preferred segregated solutions for them, also perceived them as having greater employment barriers. In the US, negative employer assumptions were also reported as frequent barriers to employment (36%) (Houtenville et al., 2015).

Negative presumptions about people with disabilities, as well as lack of consideration for job applicants with disabilities can play out as employment barriers at various stages of the application process and beyond. Evidence of discriminatory

employer attitudes was tested in a field experiment in which job applicants' cover letters disclosed either physical disability, cognitive disability, or did not indicate any form of disability (Ameri et al., 2018). Even though disability type was matched to job description so as not to interfere negatively with job responsibilities, this experiment found a 26% lower likelihood of employer interest in applicants with self-disclosed disabilities than those without. Other barriers identified in the application process have included inaccessible forms or tests, and interviews that failed to account for needed accommodations even when a disability was disclosed (McKinney & Swartz, 2021). In mixed methods research conducted by Jacob et al. (2022), it was found that both employers and coworkers expressed prejudiced attitudes as well as discrimination against employees with intellectual disabilities.

1.3. Societal-level barriers

Societal-level prejudices can manifest not only in laws but also in policies and practices that inhibit the ability of people with disabilities to participate equally in employment opportunities. For recent college graduates in Taiwan, perception of acceptance by society and the job market were viewed as the key factors explaining employment status, while neither disability type nor disability level were significant (Chen, 2015). Limited protections for prospective employees with disabilities (outside of those hired by large employers) may also reflect societal prejudices. Even when protections are in place, they can have mixed results, such as employer concerns about the ramifications of hiring a person with disabilities given regulations concerning their dismissal (Bonaccio et al., 2019; Kocman, Fischer, & Weber, 2018).

Friedman (2020) found that lower disability employment rates were significantly correlated with state disability prejudice scores. Misperceptions feed these types of bias. For example, the unmet need for workplace accommodations reported in the US (Maestas, Mullen, & Rennane, 2019) is likely exacerbated by employers failing to recognize that most employees with disabilities do not require workplace accommodations and that, when they do, their costs tend to be low (Baker et al., 2018).

Other societal-level barriers to employment for some people with disabilities include fear of loss or reduction in governmental supports (e.g., Veterans Affairs disability payments or caregiver stipends in the US, or disability pensions in Australia) if they

attain a paid position (Gong & He, 2019; Shepherd-Banigan et al., 2021). Transportation is also a major concern. In a nationally representative survey of Americans with disabilities, respondents indicated that lack of transportation was the third most common job search barrier that they encountered, following insufficient training, and employer misconceptions about their capabilities (Houtenville et al., 2015). Although prejudice and discrimination may originate in the larger national (Shaw et al., 2014) and state (Friedman, 2020) contexts within which employers are situated, they can have deleterious effects on the employment of individuals with disabilities at different stages of the employment process.

2. Materials and methods

2.1. Participant recruitment

Participants for this study were recruited from a Qualtrics opt-in Internet panel. Qualtrics and its partners use various recruitment techniques including member referrals, targeted email lists, gaming sites, customer loyalty web portals, permission-based networks, and social media. Qualtrics utilizes third party partners to design samples meant to meet recruitment quotas. Because of our interest in people with disabilities, we recruited from the general adult and the chronic conditions panels to oversample respondents with disabilities (about half of the total sample). While this generated a non-probability sample design, respondents were successfully recruited from each of the 50 states, and the resulting data were weighted to be more representative of the population (described below).

In all, $n = 6,350$ people accessed the online survey; $n = 2,252$ screened out of the survey for being under 18 or because their demographic quota was already met. The online study consent form that preceded access to the survey also specified the requirement that participants be younger than 65. This was done to limit our sample to traditionally working aged adults. Of the remaining $n = 4,098$, an additional $n = 1,062$ were screened out for failing to provide informed consent or for failing built-in survey quality checks used to detect inattentive or rushed responses. More specifically, to avoid exclusion, respondents were required to correctly answer one attention check, and responses completed in less than 2 minutes were rejected as rushed. These procedures resulted in a final analytic sample of $n = 3,021$,

of which $n = 1,491$ (49.3%) self-reported any disability. Of these, $n = 1,170$ were not currently employed.

2.2. Data collection procedures and instrument

The survey was conducted over the course of five days at the end of March and beginning of April 2019 and took an average of seven minutes to complete. Both Qualtrics and their third-party data hosts employ double opt-in verification as well as encryptions for all transmitted data, that is later destroyed. The data available to researchers contained no personally identifiable information but prospective participants were informed that there would be restricted researcher access to electronic copies, which would be securely stored, and password protected. Although the employment-related barrier questionnaire was not expected to generate any harm to prospective respondents, they were nonetheless informed that their participation was completely voluntary, that they could refuse to answer any question, and that they could withdraw their consent at any time. The primary ethical concern with the involvement of human subjects was the risk of identifying subjects based upon their responses. Stored data did not include email addresses or location (ISP address), and only summarized data is presented here. While there were no explicit benefits to participation, any small incentives or compensation for survey completion (e.g., cash, free downloads, or reward points) were provided at the discretion of Qualtrics and its partners. This study was approved by the University of New Hampshire Institutional Review Board (# 8054) after review of its data collection protocol.

The data collection instrument was based primarily on the CPS disability supplement, with additional questions drafted by the investigators and vetted by a disability and employment expert panel. The survey included a variety of demographic measures, with a focus on disability status and type, employment status and experiences, job search, and job quality. Because prior research has shown that the six disability identifiers used in many federal surveys, such as the American Community Survey, fail to identify many people with disabilities (Burkhauser, Houtenville, & Tennant, 2014), additional questions were included to capture additional types of disabilities that individuals might experience (e.g., psychiatric and learning disabilities). Specifics regarding the main dependent and independent variables are provided below.

The primary independent variable of interest, disability status, was assessed by asking up to nine

questions about sensory, ambulatory, psychiatric, and cognitive difficulties. A “yes” response to any of these is counted as having a disability:

- [Vision] *Do you have serious difficulty seeing even when wearing glasses?*
- [Hearing] *Do you have difficulty hearing?*
- [Ambulatory] *Do you have serious difficulty walking or climbing stairs?*
 - *Do you have any difficulty walking a quarter of a mile – about 3 city blocks?* (Asked of those who said no to item about stairs)
- [Physical] *Do you have any difficulty doing physical activities such as lifting, carrying, bending, or manipulating small objects?*
- [Cognitive] *Because of a physical, mental, or emotional condition, do you have serious difficulty concentrating, remembering, or making decisions?*
 - *Do you think you have a condition that makes it difficult in general for you to learn? Such conditions include attention problems (ADD), hyperactivity (ADHD), dyslexia, and others?* (Asked of those who said no to item about concentrating)
- *Do you have any emotional, psychological, or mental health conditions? These may include anxiety, depression, bipolar disorder, substance abuse, anorexia, as well as other conditions?* (Asked of those who said no to items about concentrating and learning)
- *Do you have a developmental disability or disorder? (This may include Down Syndrome, autism, or Asperger syndrome, as well as other conditions.)* (Asked of those who said no to items about concentrating, learning, or psychiatric conditions)

Finally, respondents who said no to all of the above were asked “*Do you have any other kind of disability?*” A few respondents reported disability types asked about in the above list (i.e., anxiety and learning disability) and these cases were backcoded into the correct disability category. Open-ended responses that could not be categorized by the above disability types (e.g., chronic illness) were coded as having no disability. In total, $n = 1,491$ of our final sample had a disability, or 46.6% of the weighted sample. This proportion is high by design, as our sampling quota was to have about half of our sample with a disability in order to provide ample statistical power to examine the employment barriers for this population.

The key dependent measure in this study was whether someone reported a barrier to employment. This was measured by responses to the following question asked of those who reported they were not working at the time of survey, *Do you consider any of the following to be a barrier to employment for you:*

- Lack of education or training
- Lack of transportation
- Loss of benefits or government assistance
- Need for special job features or accommodations
- Employer or coworker attitudes
- Disability or health condition
- Some other problem (if yes, please specify)

Respondents could select all that apply. Those who reported any of these barriers (including “other”) were counted as having experienced a barrier to employment. Among those who reported any type of disability, a follow-up question asked whether each reported employment barrier was “due to a health condition or disability.”

The survey instrument also captured standard demographic measures, including education (for which eight categories were presented and were collapsed into four: High school graduate or less, some college, 4-year degree, and advanced degree); total family income (seven categories were presented, with a low of under \$15,000 and a high of over \$100,000, which were collapsed into four categories: under \$30,000, between \$30,000 and under \$60,000, \$60,000 to under \$75,000, \$75,000 to under \$100,000, and everything higher); race/ethnicity, derived from asking about Hispanic ethnicity (yes/no), and race (multiple responses permitted); age (coded from year of birth), and region (Northeast, Midwest, South and West), constructed based on the state of residence identified.

2.3. *Description of analytic approach*

To address this study’s research questions, descriptive statistics and logistic regression models were conducted for each barrier type as well as for the presence of any barrier. Prior to multivariate analyses, survey data were weighted by gender, age group, household income, race, and region of the country using 2017 5-year American Community Survey estimates. The median weight was 0.72 (range 0.4 to 6.0). The unweighted study sample was disproportionately female and slightly younger than the working-age population. Hispanic or Latinx participants were also underrepresented in the unweighted

sample. The unweighted sample was similar to the overall population regarding educational attainment, region, and receipt of Social Security benefits (SSI or SSDI), though it was slightly lower on average income distribution. This may be due to our oversampling of people with disabilities, who are less likely to be employed. The weighted sample is presented throughout the rest of the findings, unless otherwise specified. The demographics of the unweighted and weighted sample are presented in Table 2.

3. Results

3.1. *Demographics*

As displayed in Table 2, 47 percent of our weighted sample reported having one or more disabilities. The most frequently reported disabilities by sample participants included those pertaining to upper body mobility (18%), cognition (17%), psychiatric state (16%), walking upstairs (13%), vision (11%), and hearing (10%). Each of the remaining disability types were reported by fewer than 10% of respondents.

3.2. *Employment participation*

In terms of employment status, 59.2% of people with disabilities were currently working, compared to 68.0% of those without a disability. Of those who were unemployed at the time of the survey, 76.0% had been employed previously and 24.0% had never been employed. Among those previously employed, 32.6% of people with disabilities were looking for work, compared to 36.8% of those without a disability. See Table 3.

3.3. *Barriers to employment by disability status*

Table 4 displays the prevalence of barriers unemployed adults faced by disability status. The vast majority (99% of those with a disability, and 90% of those without a disability) reported experiencing at least one type of employment barrier. Among people with disabilities, the most commonly cited barriers were health condition or disability (58%), lack of education or training (22%), and lack of transportation (21%). While lack of education or training (20%), and lack of transportation (19%) were also two of the top three most commonly reported barriers to employment for people without disabilities, this group also reported employer or coworker attitudes

Table 2
Sample characteristics

	<i>n</i>	Unweighted %	Weighted %
Total	3021	100.0	100.0
Gender			
Male	856	28.3	50.4
Female	2163	71.6	49.5
Refused	2	0.1	0.1
Age			
18–24	421	13.9	16.7
25–34	728	24.1	21.5
35–44	737	24.4	20.3
45–54	590	19.5	20.9
55–64	545	18.0	20.6
Race/ethnicity			
White, non-Hispanic	2098	69.5	61.4
Black, non-Hispanic	339	11.2	12.2
Other, non-Hispanic	292	9.7	8.6
Hispanic	285	9.4	17.5
Refused	7	0.2	0.2
Disability (<i>any of the 9 disability types on the survey</i>)			
Yes	1491	49.3	46.6
No	1530	50.7	53.4
Disability type			
Vision	348	11.5	11.5
Hearing	276	9.1	10.3
Walking climb stair	392	13.0	12.9
Walking quarter of a mile	173	5.7	5.3
Upper Mobility	585	19.4	18.3
Cognitive	545	18.0	16.8
Learning	182	6.0	7.6
Psychiatric (e.g., anxiety, depression)	446	14.8	16.5
IDD	8	0.3	0.5
Other	23	0.8	1.5
Education			
8th grade or less	14	0.5	0.5
Some high school	111	3.7	3.7
High school graduate or GED	688	22.8	21.9
Some college, no degree	719	23.8	23.3
Associate degree, occupational or vocational program	320	10.6	9.3
Associate degree, academic program	128	4.2	3.6
Bachelor's degree	700	23.2	24.1
Master's degree	257	8.5	9.8
Professional school degree	44	1.5	2.1
Doctorate degree	38	1.3	1.8
Income			
Less than \$15,000	390	12.9	11.5
\$15,00–29,999	499	16.5	14.7
\$30,000–44,999	474	15.7	13.4
\$45,000–59,999	383	12.7	11.1
\$60,000–74,999	324	10.7	8.9
\$75,000–\$99,999	357	11.8	11.5
\$100,000 & over	445	14.7	24.4
DK	148	4.9	4.6
Social Security (SSI or SSDI)			
Yes	367	12.2	13.0
No	2583	85.6	84.7
DK	69	2.3	2.3
Region			
Northeast	536	17.8	17.5
Midwest	668	22.2	21.1
South	1167	38.8	37.7
West	637	21.18	23.6

Table 3
Employment participation status, by disability status (weighted %; unweighted *n*)

	Total		Disability				Total		No Disability				Total	
	Looking		Not Looking		Total		Looking		Not Looking		Total		Total	
	N	%	N	%	N	%	N	%	N	%	N	%	N	%
Total	3021	100.0					1491	100.0					1530	100.0
Currently employed	1851	63.9	na	na	na	na	839	59.2	na	na	na	na	1012	68.0
Previously employed	901	76.0	161	32.6	370	67.4	531	100.0	133	36.8	237	63.2	370	100.0
Never employed	269	24.0	48	33.9	73	66.1	121	100.0	41	34.2	107	65.8	148	100.0
Missing	0													

Table 4
Prevalence of barriers faced among the unemployed, by disability status (unweighted *N*, weighted %)

	Total		Person with a Disability				No Disability	
	Barrier Faced		Attributed to Disability		Barrier Faced		Barrier Faced	
	N	%	N	%	N	%	N	%
Total	1170		652				518	
Any barriers	1116	94.7	641	98.6			475	90.4
Lack of ed or training	263	21.0	152	22.2	45	34.8	111	19.8
Lack of transportation	243	20.1	144	21.4	40	30.6	99	18.6
Loss of benefits	98	9.0	61	10.7	44	67.1	37	7.2
Need for accommodations	123	11.5	83	13.6	57	66.0	40	9.1
Employer or coworker attitudes	139	13.2	70	11.9	22	29.3	69	14.7
Disability or health	429	35.3	377	57.9	na	na	52	10.0
Other	141	9.7	60	8.3	14	18.4	81	11.4

(15%) as a barrier, more often than people with disabilities (12%). Among people with disabilities, need for accommodations ranked fourth (14%) and loss of benefits ranked sixth (11%) out of the seven barriers listed, and it is worth noting that each of these barriers were highly attributed to individuals' disabilities (66% and 67%, respectively). Respondents without disabilities were also able to cite health as a barrier. Since they did not report one of the disability types asked about in the survey, a "health" barrier would be something other than a disability, as defined here.

3.4. Barriers at different stages of job acquisition

In Table 5 the frequency of different barriers is presented relative to distinct stages in the job search process for unemployed respondents, namely looking for work, applying for work, or accepting a position. Means for the frequency with which people with and without disabilities experienced employment barriers (1 = Never to 5 = Always) are compared. Though overall means were low, indicating that experience with each individual barrier was infrequent, respondents indicated that a health condition or disability

was the most common employment barrier across all stages of the job acquisition process, with means ranging from 2.03 at the job acceptance stage to 2.32 at the job seeking and application stages. Similarly, lack of education or training and lack of transportation generally alternated between the second or third most-cited reasons across job search stages.

Beyond these general trends, however, mean comparisons between people with and without disabilities revealed significant differences ($p < 0.05$) for all barriers and job acquisition stages except the lack of education or training at the job acceptance stage. At all stages, respondents with and without disabilities ranked lack of education or training and lack of transportation among their top three most frequent employment barriers. People with disabilities identified disability or health condition as their most frequent barrier. Employer or coworker attitudes rounded out the top three barriers for people without disabilities at all job acquisition stages, though it was ranked last by people with disabilities except regarding job acceptance. People without disabilities on average also ranked loss of benefits as a more frequent barrier for them than people with disabilities across job acquisition stages. Whereas trends were

Table 5
Impact of barriers on stages of the job acquisition process among the unemployed, by disability status weighted

	Total						Disability						No Disability						Means comparison				Prob>F
	Look		Apply		Accept		Look		Apply		Accept		Look		Apply		Accept		Look	Apply	Accept		
	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd					
Lack of education or training	2.06	1.16	2.19	1.20	1.66	1.01	2.16	1.19	2.34	1.24	1.68	1.03	1.92	1.11	2.01	1.11	1.63	0.98	0.000	0.000	0.388		
Lack of transportation	2.11	1.27	2.08	1.29	1.87	1.21	2.27	1.32	2.24	1.33	2.02	1.29	1.90	1.17	1.88	1.20	1.68	1.07	0.000	0.000	0.000		
Loss of benefits	1.75	1.17	1.74	1.19	1.59	1.08	1.89	1.26	1.86	1.29	1.66	1.16	1.58	1.01	1.58	1.04	1.51	0.97	0.000	0.000	0.011		
Need for accommodations	1.81	1.16	1.80	1.17	1.68	1.10	2.02	1.27	2.01	1.29	1.83	1.22	1.54	0.95	1.54	0.93	1.50	0.91	0.000	0.000	0.000		
Employer or coworker attitudes	1.74	1.05	1.74	1.04	1.71	1.03	1.85	1.11	1.84	1.12	1.78	1.09	1.61	0.95	1.62	0.91	1.61	0.94	0.000	0.000	0.004		
Disability or health	2.32	1.49	2.32	1.49	2.03	1.41	2.99	1.52	2.96	1.52	2.52	1.54	1.49	0.94	1.50	0.95	1.42	0.90	0.000	0.000	0.000		
Other	not asked																						

Scale for Barrier Frequency Survey Questions: 1 = Never 2 = Rarely 3 = Sometimes 4 = Frequently 5 = Always.

consistent across job acquisition stages for people without disabilities, the frequency of different barriers shifted for people with disabilities at the job acceptance stage. At this stage, lack of education or training fell to fifth, and need for accommodations increased, while the loss of benefits as an employment barrier declined for people with disabilities.

3.5. Demographic and socioeconomic predictors of employment barriers

Based on logistic regressions conducted on demographic variables for the presence of each individual employment barrier, as well as the presence of any barrier at all, it was observed that only the presence of a disability significantly increased the likelihood of an unemployed person reporting an employment barrier (analysis not shown). Given these results, logistic regressions were rerun replacing the presence of any disability variable with dichotomous dummy variables for each disability type. For this analysis cognitive, learning, and intellectual or developmental disabilities were combined, as were the physical challenges of climbing stairs, walking a quarter mile or upper body mobility difficulties.

Results of this analysis showed that experiencing any employment barrier was significantly associated with disability status, family income, age, and educational attainment (see Table 6). Again, disability was the predictor most significantly associated with increased risk of experiencing an employment barrier at all. Of these, having a *cognitive, learning, or intellectual disability* most significantly increased the likelihood of reporting any employment barrier. The only unique employment barrier significantly associated with this likelihood was one's disability or health condition. For people with *physical disabilities*, the risk of experiencing some employment barriers increased significantly in terms of the need for accommodations, loss of benefits, or barriers because of their disability or health condition, relative to people without physical disabilities. They were significantly less at risk however for lack of transportation, or other barriers. People with *psychiatric disabilities* experienced significantly greater employment barrier risks regarding their disability or a health condition, but significantly less risk regarding need for accommodation or loss of benefits than those without psychiatric disabilities. Notably, people with vision disabilities were significantly less likely than those without a *vision disability* to report coworker attitudes as an employment barrier than those without

vision disabilities. No specific employment barriers were significantly associated with having a *hearing disability*.

Although there was some variation in non-disability/health condition predictors by specific barrier type, family income, age, and gender were significantly associated with different types of barriers most often. Being *female* significantly reduced one's odds of experiencing employment barriers related to having a disability or health condition, or a need for an accommodation, but increased the odds of experiencing other barriers. Older age also significantly reduced one's chances of experiencing barriers related to need for accommodation, or insufficient education or training, but increased once chance of having a disability or health condition serve as an employment barrier, compared to younger adults. In terms of *annual family income*, significant differences were only found between families earning \$60,000 or more and families earning less than \$30,000. Families earning between \$30,000 and \$59,999 were not significantly different in terms of employment barriers to families earning less than \$30,000. For families with incomes of \$60,000 or more, transportation was less likely to be an employment barrier. Only families with incomes between \$60,000 and \$74,999 were significantly less likely to report disability or health condition as an employment barrier, compared to those earning less than \$30,000 a year. For families with incomes of \$75,000–\$99,999, "other" employment barriers were significantly more likely to be an issue than for those earning less than \$30,000 a year, although the opposite was the case for those earning \$100,000 or more.

3.6. *Predicting attribution of other barriers to disability or health*

In this study, survey respondents were not only asked to identify direct employment barriers, but also to indicate whether a barrier may be attributable to a disability or health condition that they have. Because attribution for each non-disability or health condition was at least partially attributed to one's disability or health condition (see Table 4), logistic regressions were run to identify the demographic and socio-economic predictors of employment barriers for the subset of respondents who identified themselves as having a disability. Because of an insufficient number of cases, loss of benefits ($n=53$) and "other" employment barriers ($n=33$), were excluded from this analysis. (See Table 7).

Results for this group of respondents with disabilities revealed sex, education, and family income were significantly associated with the likelihood of reporting any employment barrier attributed to one's disability or health condition. Looking across employment barrier types in our analysis, however, of these only the educational attainment barrier was significantly attributed to one's disability or health condition. In terms of specific barriers, coworker attitudes as a barrier were significantly more likely to be attributed to a disability or health condition only among people holding advanced degrees, compared to those with a high school diploma or less. Similarly, the need for accommodation was attributed to a disability or health condition more often among people with a bachelor's degree, compared to a high school diploma or less.

While the need for accommodations attributed to a disability was significantly associated with the majority of demographic variables in this analysis, the nature of the relation was not uniform. While age was conversely associated with the likelihood of attributing the need for accommodation to a disability or health condition, people with disabilities who were Black and non-Hispanic were significantly less likely to attribute the need for accommodation to their disability or health condition, than White, non-Hispanic people with disabilities. People with disabilities residing in the Midwest or Northeast were also significantly less likely to attribute the need for accommodation to their disability or health condition than people with disabilities residing in the West.

3.7. *Employment barriers by educational attainment*

Because of the significance of educational attainment in this study, further analysis was conducted to look for associations between employment barriers of unemployed study participants and their levels of educational attainment. Results displayed in Table 8 indicate that there were significantly inverse relations between educational attainment and employment barriers relating to lack of education or training, lack of transportation, and disability or health condition.

4. Discussion

This study's findings echo prior research on disability employment, while also adding new information. The majority of the unemployed experienced

Table 6
Odds Ratios (and standard errors) predicting barriers to employment (any attribution), by disability type

	Any barrier	Education/ Training	Transportation	Loss of benefits	Need for accommodations	Attitudes of coworkers	Disability	Other barriers
Female (ref = male)	0.607 (0.274)	1.311 (0.262)	0.972 (0.187)	0.928 (0.251)	0.557 (0.133)*	0.654 (0.147)	0.624 (0.128)*	3.018 (0.929)***
Race (ref = White, Non-Hispanic)								
Black, Non-Hispanic	1.278 (0.895)	1.324 (0.344)	1.312 (0.343)	1.011 (0.388)	1.577 (0.539)	1.173 (0.381)	0.874 (0.263)	0.187 (0.090)**
Other, Non-Hispanic	1.169 (0.673)	1.280 (0.372)	1.019 (0.315)	0.607 (0.354)	1.630 (0.632)	1.689 (0.587)	0.907 (0.307)	0.369 (0.185)*
Hispanic	0.779 (0.545)	1.061 (0.311)	1.024 (0.301)	1.207 (0.461)	0.619 (0.287)	1.037 (0.430)	0.846 (0.298)	1.007 (0.387)
Education (ref = HS or less)								
Some college	0.915 (0.594)	1.209 (0.246)	0.783 (0.159)	1.212 (0.380)	1.539 (0.438)	0.755 (0.200)	0.950 (0.200)	0.898 (0.234)
Bachelors	0.246 (0.151)*	0.923 (0.269)	0.524 (0.174)	1.598 (0.643)	1.700 (0.690)	1.073 (0.356)	0.581 (0.178)	1.521 (0.582)
Advanced degree	0.902 (0.709)	0.280 (0.175)*	0.718 (0.414)	1.671 (1.006)	5.249 (2.970)**	1.867 (0.820)	0.454 (0.244)	0.874 (0.496)
Family income (ref = <\$30k)								
\$30k-59,999	0.809 (0.562)	1.018 (0.222)	0.966 (0.218)	0.681 (0.226)	1.270 (0.372)	1.691 (0.487)	0.698 (0.166)	0.763 (0.211)
\$60k-74,999	0.468 (0.299)	0.733 (0.257)	0.470 (0.165)*	0.452 (0.241)	0.891 (0.419)	1.397 (0.544)	0.493 (0.174)*	0.869 (0.343)
\$75k-99,999	0.624 (0.411)	1.022 (0.341)	0.417 (0.167)*	0.273 (0.227)	1.358 (0.600)	1.477 (0.609)	0.545 (0.201)	2.453 (0.850)**
\$100k+	0.214 (0.134)*	0.681 (0.246)	0.321 (0.147)*	1.096 (0.445)	1.130 (0.511)	0.880 (0.347)	0.706 (0.242)	0.320 (0.145)*
Age	0.969 (0.016)*	0.977 (0.006)***	0.988 (0.006)	0.988 (0.010)	0.963 (0.010)***	0.991 (0.009)	1.048 (0.008)***	1.002 (0.009)
Region (ref = West)								
Midwest	0.837 (0.421)	0.777 (0.216)	0.905 (0.248)	1.476 (0.606)	0.832 (0.305)	0.856 (0.285)	1.589 (0.425)	0.757 (0.229)
South	1.946 (1.165)	0.761 (0.179)	0.805 (0.199)	1.234 (0.432)	1.010 (0.323)	1.005 (0.301)	1.617 (0.421)	0.722 (0.212)
Northeast	1.438 (0.815)	0.964 (0.280)	0.646 (0.195)	1.801 (0.834)	0.854 (0.363)	0.880 (0.360)	1.794 (0.537)	0.408 (0.149)*
Cognitive/learning/intellectual difficulty	15.885 (17.421)*	1.419 (0.343)	1.488 (0.373)	1.156 (0.355)	1.249 (0.373)	1.440 (0.399)	7.506 (1.773)***	0.879 (0.308)
Physical difficulty	11.469 (9.528)*	0.686 (0.150)	0.523 (0.121)**	2.126 (0.749)*	3.598 (1.098)***	0.825 (0.237)	5.658 (1.150)***	0.395 (0.136)***
Emotional/ psychological/ mental health difficulty	2.750 (1.621)	1.631 (0.465)	1.659 (0.438)	0.364 (0.183)*	0.418 (0.175)*	0.951 (0.379)	4.541 (1.128)***	1.512 (0.458)
Hearing difficulty	0.234 (0.142)*	1.349 (0.460)	1.023 (0.323)	1.468 (0.536)	1.541 (0.580)	1.585 (0.571)	0.869 (0.296)	0.244 (0.192)
Vision difficulty	1.000 (omitted)	0.577 (0.215)	1.284 (0.378)	1.219 (0.500)	0.748 (0.282)	0.229 (0.112)**	1.032 (0.311)	1.963 (0.909)***
_cons	145.171 (200.925)***	0.665 (0.272)	0.735 (0.289)	0.092 (0.048)***	0.279 (0.142)*	0.236 (0.134)*	0.022 (0.011)***	0.103 (0.048)***

* $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

Table 7

Odds Ratios (and standard errors) predicting barriers to employment attributed to disability/health condition (includes only those coded as having a disability)

	Any barrier attributed to disability/health condition	Education/ Training	Transportation	Need for accommodations	Attitudes of coworkers
Female (ref = male)	0.488 (0.119)**	0.362 (0.189)	0.378 (0.205)	1.572 (1.517)	2.710 (1.862)
Race (ref = White, Non-Hispanic)					
Black, Non-Hispanic	0.470 (0.191)	1.385 (0.762)	0.562 (0.420)	0.078 (0.079)*	0.801 (0.963)
Other, Non-Hispanic	0.659 (0.346)	1.653 (1.569)	0.581 (0.581)	0.925 (0.959)	0.552 (0.670)
Hispanic	0.625 (0.285)	2.086 (0.140)	0.948 (0.919)	0.049 (0.080)	1.004 (1.466)
Education (ref = HS or less)					
Some college	1.716 (0.470)*	1.089 (0.586)	0.839 (0.422)	6.500 (6.132)	0.721 (0.552)
Bachelors	1.144 (0.506)	1.219 (1.146)	0.679 (0.586)	11.964 (13.239)*	1.244 (1.266)
Advanced degree	1.722 (1.224)	0.885 (1.456)	—	—	21.214 (29.860)*
Family income (ref = <\$30k)					
\$30k-59,999	0.744 (0.235)	0.735 (0.401)	1.137 (0.685)	0.379 (0.324)	0.240 (0.181)
\$60k-74,999	0.307 (0.165)*	0.677 (0.777)	—	0.078 (0.105)	0.103 (0.201)
\$75k-99,999	0.465 (0.228)	0.236 (0.234)	0.959 (0.896)	0.203 (0.364)	1.154 (1.131)
\$100k+	1.180 (0.616)	0.361 (0.356)	4.923 (5.597)	—	0.452 (0.555)
Age	0.992 (0.009)	1.007 (0.017)	0.993 (0.018)	1.054 (0.026)*	1.011 (0.026)
Region (ref = West)					
Midwest	1.036 (0.357)	2.151 (1.616)	1.254 (0.918)	0.097 (0.163)*	1.721 (2.067)
South	1.041 (0.343)	2.434 (1.690)	0.973 (0.639)	0.076 (0.121)	1.008 (1.090)
Northeast	0.621 (0.243)	0.528 (0.496)	0.384 (0.367)	0.028 (0.048)*	0.449 (0.671)
_cons	0.710 (0.352)	0.530 (0.614)	1.242 (1.262)	1.897 (3.502)	0.283 (0.391)
Number of cases	596	142	130	70	67

* $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

Table 8

Percentage of those not working who reported barriers to employment, by level of education (values are row %) - Weighted N

	Lack of education	Lack of transportation	Loss of benefits	Need for accommodation	Employer/ coworker attitudes	Disability	Other
High school or less	22.8%	26.7%	7.8%	10.4%	13.8%	39.3%	8.4%
Some college/Associate's	24.0%	19.1%	9.6%	11.2%	10.7%	38.6%	10.5%
Bachelor's	17.6%	11.1%	9.3%	10.0%	14.4%	28.2%	12.8%
Advanced degree	5.9%	10.3%	11.9%	21.7%	19.3%	13.6%	6.5%
<i>Chi Square</i>	16.8785**	28.0118**	1.9364	10.2575	5.6442	28.1671***	4.3965

similar employment barriers, but people with disabilities reported disability or health condition most often, and people without disabilities reported organizational attitudes, and other barriers more often. In general, people with physical mobility, learning-related, or psychological limitations experienced significantly more employment barriers than people with vision or hearing limitations, compared to those with no disabilities. Statistically significant differences were found between people with and without disabilities across barrier types, except regarding a lack of education or training, at the job acceptance stage of the job acquisition process.

While previous research led us to expect particular employment barriers among survey respondents with self-reported disabilities, such as lack of qualifications or inaccessible transport (Devine et al.,

2021), one surprise was the similarly high prevalence of job seekers without disabilities who experienced any job search barriers (90%). Even though people with and without disabilities report encountering some of the same types of barriers to employment (Phillips et al., 2020), our finding that people with disabilities experienced barriers significantly more frequently when searching for work was also somewhat expected (Meltzer, Robinson, & Fisher, 2020; McKinney & Swartz, 2021).

All respondents, regardless of disability status, cited a lack of education or training and lack of transportation as among the most common barriers. As in past studies, people with disabilities were most likely to identify their disability or health condition as the major barrier to employment. An unexpected finding was that respondents without dis-

abilities ranked employer or coworker attitudes as a *more frequent* employment barrier than people with disabilities. This may reflect other social identities among participants without disabilities that were not explored in depth in our study. For example, L'Horty et al. (2022) have found that racial and ethnic discrimination can occur to a similar or even greater extent than disability-related discrimination in some occupations. Although we did not find racial identity to be significantly associated with employer or coworker attitudes directly, our finding that Black, non-Hispanic people with disabilities were significantly less likely to attribute accommodation barriers to their disability implies a need for further investigation of racial disparities in disability employment by occupation. Moreover, because of the lack of a statistically significant association between employer or coworker attitudes and gender, education, income, or age, as well as race or ethnicity for those without disabilities, these attitudinal barriers to employment may instead be based on other marginalized identities. In addition, it may be the co-existence of multiple prejudicial attitudes that, as was reported in a scoping review on employment barriers for individuals with disabilities, be particularly problematic, as was observed for veterans with disabilities (Shaw et al., 2014). Considering these unobserved possibilities, it is therefore suggested that future research efforts should incorporate other identities, such as immigration status (Keita & Valette, 2019), sexual orientation (Fric, 2021), and veteran status in examinations of employment barriers for individuals with disabilities that have been found to be meaningful in the past but that were not included in our survey.

Another potentially surprising finding, that runs counter to some of the economic literature (Barlev-Kotler, Rivkin, & Sandler-Loeff, 2014; Brew & Gleason, 2014), was the low ranking of loss of benefits as a barrier among people with disabilities. Such concerns may be reflected in the extreme hesitancy (only 1%) of working aged Israelis with disabilities to increase their income beyond a prior disability benefit income cut-off point even three years after the allowance amount was lifted (Barlev-Kotler, Rivkin, & Sandler-Loeff, 2014).

Meanwhile in the U.S., Brew and Gleason calculated that only 0.2% of people returned to work once they started receiving SSI and SSDI benefits. Providing greater context to these reports however, Yosef, Soffer, and Malul (2019) demonstrated that individuals with disabilities were actually more likely to prefer to maintain whatever their current income

source was regardless of whether it was employment or welfare, than to change it.

In our study, loss of benefits was infrequently mentioned (10.7%), ranking sixth of the seven barriers listed. Anand and Sevak (2017) observed a similarly low ranking of benefits loss as an employment barrier among nonworking individuals with disabilities, although at a higher percentage (23%). This does not mean that loss of benefits is not a legitimate concern, but only that the other employment barriers listed were considered more problematic for survey respondents. Its rank below disability, education and training, lack of transportation, need for accommodations, and employer or coworker attitudes however is consistent with previous U.S. Bureau of Labor Statistics findings in 2019 (U.S. Bureau of Labor Statistics, 2021). Moreover, loss of benefits was reported as a barrier by about 7% of people without disabilities, who may have been receiving benefits for income, as opposed to disability-related reasons. Interestingly, the loss of benefits employment barrier became less frequent across the different job acquisition stages for those with and without disabilities. This is consistent with prior research indicating that fear of benefit loss can even discourage individuals from seeking employment-related services (Shepherd-Banigan et al., 2021). Loss of benefits was also significantly more problematic for people with physical disabilities and least significant for people with psychiatric disabilities. Taken together our findings regarding loss of benefits argues for future qualitative research to help explain the greater reports of this concern among those with physical disabilities than those with psychiatric disabilities as well as policy interventions such as New York state's six-month, 100% income disregard for residents receiving public assistance, based on a successful New York City pilot study (Duffy & Seeley, 2023).

Because this study was conducted prior to the COVID-19 pandemic we expect that the need for accommodations as an employment barrier may have changed for prospective employees both with and without disabilities. This supposition is supported by a 2022 survey of workplace supervisors, 78% of whom stated that their organization established or changed their accommodation process due to COVID-19-related issues (Houtenville et al., 2022). More specifically, among participating supervisors, 47% indicated a moderate or large increase in the percentage of paid employees working from home. Sixty-nine percent of those reporting this increase also indicated that they anticipated the per-

centage of those engaged in telework to continue growing.

Given the historically low unemployment rate in the US (Butera, 2023), and because the need for accommodations employment barrier was most problematic at the job acceptance stage, it is likely that it would be perceived as less of a barrier now than prior to the pandemic. Telework, in response to America's current labor shortage (Ferguson, 2023), is also likely to reduce the lack of transportation that people with and without disabilities ranked as one of their top three employment barriers.

Nevertheless, teleworking may not be a universal solution. As a recent pre-post difference-in-difference analysis revealed, even when rural community members were provided access to a wireless broadband network, it did not appear to alter their educational or employment-related outcomes (Valentin-Sivico et al., 2023). Therefore, as was suggested by a recent scoping review (Fortune et al., 2022), in order to uncover the differential use and benefits of telework interventions, it may first be necessary to pursue qualitative research to more comprehensively appreciate how and for whom they are more or less effective.

Given that telework will not remediate all transportation-related employment barriers for individuals with disabilities, other strategies must be enlisted, such as transportation accommodations, which have been positively correlated with current employment for people with either multiple disabilities or exclusively physical disabilities in the past (Anand & Sevak, 2017). Because transportation barriers vary and may include lack of availability, challenges commuting when transportation is available, and other issues (Inge et al., 2018), transportation solutions will need to account for contextual factors. For instance, interventions to improve transportation options for employment-seeking people with disabilities in rural areas, such as the "Automated Driving Systems for Rural America" which was positively received by its older adult participants (Mason, Carney, & Gaspar, 2022), may be sensitive to population density differences.

Because e-learning and distance learning has however been shown to be beneficial for some individuals with learning disabilities (Petretto et al., 2021), its expanded use since the onset of the pandemic may similarly have reduced some of their educational or training barriers. The significance of education on employment continually emerged across study findings but showed variation by job acquisition phase

and barrier type. Similar to prior research, we found that insufficient education or training was considered a primary employment barrier for people with disabilities (Sundar et al., 2018). This is not surprising, given that people with disabilities with college degrees are more likely to have acquired a two-year or community college degree, which is less likely to be associated with post-collegiate employment (Phillips et al., 2022). Regardless of disability status, lower levels of education were significantly associated with educational, transportation, disability, or health condition-related employment barriers. Future interventions and research might seek to differentiate between the benefits of academic and vocational training programs as a recent longitudinal study in Australia has shown that graduates of apprenticeship and training programs achieved and sustained positive employment outcomes (Thoresen, Cocks, & Parsons, 2021).

Although we found significant differences in the extent to which insufficient education or training was an employment barrier between people with and without disabilities during job acquisition stages, it was not significant at the job acceptance stage. Insufficient education or training was most significant at the application stage, which indicates that training, policies, and interventions targeting this aspect of job seeking might be particularly beneficial for people with disabilities across disability types, as well as for people without disabilities.

Finally, where disability type was a significant factor in the experience of job acquisition challenges, people with intellectual, cognitive, or psychiatric disabilities experienced barriers most frequently. This finding suggests the need for future interventions directed at improving the employment outcomes for members of these subpopulations, who, as previously indicated, may be at greater risk of unemployment (Kocman, Fisher, & Weber, 2018; McAlpine & Alang, 2021). Successful efforts such as Mentra's online job matching website for neurodivergent individuals may for example provide a useful model for individuals with different mental health challenges (Peluso, 2023).

Other significant differences by disability type were also identified. People with vision disabilities were significantly less likely than people with other types of disabilities to report negative attitudes as a barrier. Although negative employer attitudes had previously been ranked as a greater barrier by college graduates with visual impairments seeking employment (Antonelli, Steverson, & O'Mally, 2018), our

findings may differ because they controlled for educational attainment. At the same time, nonworking individuals with advanced degrees were more likely to perceive accommodation needs as a barrier. Further qualitative research may help explain this finding as both those with and without disabilities described lack of accommodation as a barrier in our study, and those with advanced degrees represent only 7.8 and 15.1%, respectively of those 25 or older in the civilian population of the US in 2021 (Paul et al., 2023).

Despite the contributions of this study, some limitations must be acknowledged. First, our survey was based on self-reported perceptions of barriers by nonworking individuals with and without disabilities. Unfortunately, more objective measures of the range of impediments considered here would be difficult to measure in alternative ways. Reviewing reports of hiring discrimination for instance would likely undercount discouraged and underemployed job seekers. In addition, the percentage of the sample with disabilities that is employed is high (59%), which may be a result of the expanded definition of disability (capturing 9 disability types instead of the 6 more commonly used). We chose this approach in hopes of disaggregating cognitive disability and investigating differences among some of the supplemental types of disabilities. Unfortunately, sample sizes did not always permit these analyses. Another limitation is that the disability question skip patterns we used may have underrepresented people with concentration, memory, or decision-making disabilities who also have learning, psychiatric, and developmental disabilities.

The Qualtrics online sample is also biased towards people who have internet access. People with disabilities are overrepresented among those lacking internet access, which can be an employment barrier as well as a barrier to study participation. Although the Qualtrics Internet panel was derived from a non-probability sample design, the researchers applied weights for gender, age group, household income, race, and region of the country using 2017 5-year American Community Survey estimates to provide more representative results.

The survey instrument adapts survey questions that have been developed by federal statistical agencies, including questions on disability and barriers to employment present on the Current Population Survey-Disability Supplement. This approach allowed us to benefit from the methodological work of the designers of such national studies, although we did not specifically test for validity of the specific survey instrument used in this study.

Lastly, although our conceptualization of employment barrier levels was based on a prior conceptualization (Sundar et al., 2018), there can be overlap between categories. Employer and coworker attitudes may reside within the individual or the corporate culture of a particular organization, for example. Situating employer and coworker attitudes within the workplace, however, may be particularly meaningful in explaining one's overall success with job acquisition and be a useful basis of comparison in regard to retention.

5. Conclusion

This study assessed individual-, organizational-, and societal-level employment barriers faced by people with and without disabilities across different stages of the job search process. The results indicate that employment barriers faced by people with and without disabilities vary less in nature than by degree. Thus, improvements in transportation, and better alignment between education and training and the job market may benefit all prospective employees, as well as employers. Because people with disabilities still experience more employment barriers, additional efforts targeting recruitment, the application process, and job negotiations are warranted. In light of the novel finding that insufficient training was seen most frequently as a barrier at the application stage for people with and without disabilities, this is a ripe area for more attention by researchers and employment service professionals, such as vocational rehabilitation, to address challenges at this particular stage.

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None to report.

Conflict of interest

The authors declare that they have no conflicts of interest.

Ethics statement

The Institutional Review Board for the Protection of Human Subjects in Research (IRB) at the Univer-

sity of New Hampshire has reviewed and approved the protocol for this study (IRB # 8054).

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Informed consent

All participation was based on informed consent.

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